

Social associations across species during nocturnal bird migration

Benjamin M. Van Doren Joely DeSimone Josh A. Firth
Friederike Hillemann Zach Gayk Emily Cohen
Andrew Farnsworth

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Summary

An emerging frontier in ecology explores how organisms integrate social information into movement behavior and the extent to which information exchange occurs across species boundaries (Flack et al. 2022; Aikens et al. 2022; Oestreich et al. 2022). Most migratory landbirds are thought to undertake nocturnal migratory flights independently, guided by endogenous programs and individual experience (Newton 2008; Åkesson and Helm 2020). Little research has addressed the potential for social information exchange aloft during nocturnal migration, but social influences that aid navigation, orientation, or survival could be valuable during high-risk migration periods (Flack et al. 2022; Aikens et al. 2022; Loonstra et al. 2023; Byholm et al. 2022; Dhanjal-Adams et al. 2018). We captured audio of >18,000 h of nocturnal bird migration and used deep learning to extract >175,000 in-flight vocalizations of 27 species of North American landbirds (Hamilton 1962; W. R. Evans and O'Brien 2002; Farnsworth 2005; Van Doren et al. 2024). We used vocalizations to test whether migrating birds distribute non-randomly relative to other species in flight, accounting for migration phenology, geography, and other non-social factors. We found that migrants engaged in distinct associations with an average of 2.7 ± 1.9 SD other species. Social associations were stronger among species with similar wing morphologies and vocalizations. These results suggest that vocal signals maintain in-flight associations that are structured by flight speed and behavior (Gayk, Simpson, and Mennill 2021; Gayk and Mennill 2023; Farnsworth 2005). For small-bodied and short-lived bird species, transient social associations could play an important role in migratory decision-making by supplementing endogenous or experiential information sources (Németh and Moore 2007; Larkin and Szafoni 2008; Cohen and Satterfield 2020). This research provides the first quantitative evidence of interspecific social associations during nocturnal bird migration, supporting recent calls to rethink songbird migration with a social lens (Aikens et al. 2022). Substantial recent declines in bird populations (Rosenberg et al. 2019; Bairlein 2016) may diminish the frequency and strength of social associations during migration, with currently unknown consequences for populations.

Results

The migratory journeys of diverse taxa frequently overlap in space and time (Cohen and Satterfield 2020). Bird migration is a prime example, with hundreds of millions of individuals of dozens of species often in the air on a given night (Cohen and Satterfield 2020; Van Doren and Horton 2018). Though existing research on navigation and decision-making during songbird migration has often emphasized the role of endogenous timing and navigation programs, opportunities for social information exchange occur frequently during stopover (Németh and Moore 2014, 2007; Rodewald and Brittingham 2002; DeSimone et al. 2024) and migratory flight, when many taxa actively vocalize (Farnsworth 2005; W. R. Evans and O'Brien 2002; Morris et al. 2016). In-flight vocalizations may be important for communicating social information en route (Gayk, Simpson, and Mennill 2021; Gayk and Mennill 2023; Farnsworth 2005;

Griffiths et al. 2016; Ress et al. 2024; Winger et al. 2019), and social information could aid in navigational decision-making, finding appropriate stopover habitat, or identifying other individuals with which to form mixed-species foraging flocks during stopover. In other contexts, social information use among heterospecifics has been demonstrated empirically (e.g., nest site choice (Seppänen et al. 2007); terrestrial migration stopover (Németh and Moore 2007); foraging (Hillemann et al. 2020; Firth, Sheldon, and Farine 2016)). However, it is unknown what information might be exchanged or used during nocturnal migratory flights (Cohen and Satterfield 2020).

Here, we investigate whether nocturnally migrating bird species form consistent social associations during migratory flights. We use recordings of in-flight vocalizations to characterize patterns of species’ spatial and temporal proximity and test whether species’ distributions aloft differ from a null expectation based on non-social factors including shared phenology and geography. Significant differences from the null hypothesis would suggest an active behavior driving social association among species. We investigate the factors that explain any species associations, hypothesizing that species with similar migration routes, stopover habitats, morphologies, vocalizations, and evolutionary histories will be more likely to socially associate. Finally, we consider how social information exchange could be an important contributor to the movement ecology of nocturnally migrating birds.

Songbirds associate with other species during migratory flights

We collected acoustic recordings of 18308.08 hours of autumn nocturnal bird migration (August to December) from 26 sites in eastern North America. We extracted vocalizations of migrating birds (hereafter “flight calls” (W. R. Evans and Rosenberg 2000; Farnsworth 2005)) from audio data using a deep learning model that we designed for this purpose (Van Doren et al. 2024), and we manually reviewed species detections for accuracy to ensure data quality. We focused on 27 well-sampled species: 25 songbird species (Order Passeriformes), plus two heron species (Order Pelecaniformes), which we included to examine the potential for associations between songbirds and other orders. We constructed a network that captures the degree to which detections of different species occurred synchronously in the data stream conditional on species co-occurrence (hereafter “social association network”). Using custom network permutation tests, we evaluated whether the observed social association network differed from a null expectation that incorporated shared timing, geography, and other non-social factors that may contribute to network structure. The observed social association network was significantly non-random (network coefficient of variation $P_{30s} = 0$). We quantified the overall tendency of each species to associate with other species during migratory flights, finding that 17 out of 27 species in the social association network showed significantly elevated total association strengths after accounting for non-social factors (Figure 1). For this study, we considered detections to occur synchronously if they occurred in the same 30-s time window, but we also tested networks constructed using 15-s and 60-s time windows and confirmed that the results

were robust to the choice of window size (15-s social association: 17 of 27 significant; 60-s social association: 20 of 27 significant).

We assessed the statistical significance of social association for every species pair in the network using custom permutation tests that accounted for non-social factors that may contribute to network structure. Of 213 species pairs with >100 association opportunities assessed using 30-s time windows, 36 were statistically significant (Figure 2). This result was consistent when using other window sizes (15-s: 35/215 pairs significant; 60-s: 35/210 pairs significant). For 30-s windows, species had a mean of 2.7 ± 1.9 SD significant association partners, and 23 of 36 significant associations were between two species of the same family (most commonly within the Parulidae). Although significant interfamilial associations were less frequent, those that did occur were of similar strength to intrafamilial associations (t -test: $t = 0.4$, $df = 21.5$, $P = 0.69$).

Wing morphology and vocalization similarity explain social associations among species

We tested whether in-flight associations among species could be explained by phylogeny, spatiotemporal distribution, habitat preferences, social relationships during stopover, morphology, or vocalizations. We used non-parametric Mantel tests and again evaluated statistical significance using custom network permutations that accounted for non-social factors. The similarity of species' wing lengths and their vocalizations were statistically significant predictors of social association (Figure 3; Table 1). These relationships were robust to choice of window size and present using parametric and nonparametric matrix correlations. These relationships were also present when excluding two large-bodied heron species and including only species in the order Passeriformes. Species relationships at stopover, phylogenetic relatedness, spatiotemporal overlap in species' migration routes, non-breeding range overlap, and migration-period habitat relationships were not consistently associated with social association (Table 1), although stopover affiliation index, migration overlap, and non-breeding range overlap showed some support at other window sizes.

Discussion

Little is presently known about how organisms integrate interspecific information into behavioral decision-making, a topic at the cutting edge of ecology (Aikens et al. 2022; Flack et al. 2022; Cohen and Satterfield 2020; Piersma 2022). Songbirds migrate primarily at night and are typically thought to do so independently, without information contributions from other birds (Åkesson and Helm 2020). However, our results demonstrate that songbirds engage in interspecific social associations during nocturnal migratory flights. The majority of bird species studied showed significantly higher association strengths than expected under null models accounting for species co-occurrence and non-social factors, indicating that it is more likely for

these species to occur with heterospecifics than expected by chance. Social associations were most frequent among species of the same family, particularly wood-warblers in the family Parulidae, but significant interfamilial associations were also frequent and no less strong when present. In contrast, we did not find strong evidence of social associations across orders (e.g., between Passeriformes and Pelecaniformes).

Stronger social associations tended to occur between bird species with more similar wing lengths, but not closer phylogenetic relatedness, suggesting that flight speed may be important in structuring in-flight associations (Wang, McGowan, and Dyke 2011). Over the course of hours-long migratory flights, individuals with similar flight speeds and altitudes may more easily maintain close proximity and sustain an association, whereas individuals with different flight behaviors are more likely to grow gradually apart, making any such associations ephemeral. Associations were also stronger among species with more similar vocalizations, a finding consistent with the hypothesis that flight calls are used to maintain multi-species associations during migratory flights (Hamilton 1962; Gayk and Mennill 2023). These findings suggest the possibility that shared migratory behavior may be driving convergent evolution in acoustic signals across species (Gayk, Simpson, and Mennill 2021).

In contrast, we found no consistent evidence that fine-scale in-flight associations were linked to habitat preferences, geographic ranges, or species affiliations during diurnal stopovers. This result is surprising given evidence for heterospecific attraction in Palearctic birds, where playback of heterospecific vocalizations can cause migrating birds to land and initiate stopover (Hera et al. 2017). Our results suggest that the interspecific relationships among migrants in the Americas re-shuffle as they alternate between nocturnal aerial and diurnal terrestrial habitats, with variables related to flight behavior shaping in-flight relationships and variables related to foraging behavior shaping stopover relationships (DeSimone et al. 2024). Previous work has connected vocalization similarity to migration range overlap (Gayk, Simpson, and Mennill 2021), but the lack of an association between geographic range and social association in our data suggests that spatial overlap may not be the primary driver of associations aloft. Although we did not detect a strong spatial signal in our data, we advocate for exploring the how social associations may vary across space and how an individual’s social behavior is influenced by its spatial-social context (Webber et al. 2023).

The importance of social information during migration

Our study provides important context for a growing body of evidence that the social information available to an individual may be an important and underappreciated contributor to migratory behavior (Oestreich et al. 2022; Aikens et al. 2022). The associations and vocalizations we detect during nocturnal bird migration may provide a conduit for information exchange. The use of social information during migration is well documented in some bird species, such as large-bodied cranes (Gruidae) and storks (Ciconiidae) (Mueller et al. 2013; Abrahms et al. 2021; Aikens et al. 2024), as well as other species that commonly form groups

or flocks, such as terns (Laridae) and shorebirds (Charadriiformes) (Byholm et al. 2022; Loonstra et al. 2023). In these species, conspecific social information is thought to be of particular importance for younger birds undertaking their first migrations.

Our results provide evidence that social information could also be transmitted during migration among small-bodied and short-lived bird species that are generally thought to undertake nocturnal migration independently (Åkesson and Helm 2020). Since these species do not learn their migration routes from their parents, social information could play an important role in supplementing information from the innate migratory program, especially for inexperienced birds. Such information could aid navigation, as has been demonstrated in large-bodied diurnal migrants, or be associated with habitat selection, stopover, or other factors. Flight calls may encode information about an individual, such as age and sex, as well as individual identity, which may allow birds to infer the flight direction of other individuals and facilitate the maintenance of group cohesion among both conspecifics and heterospecifics (Griffiths et al. 2016; Hamilton 1962; Ress et al. 2024; Larkin and Szafoni 2008).

Social behavior and information use exchange could take several forms, and our results provide a foundation for testing hypotheses about social influences on migratory behavior. We highlight several questions for future research, adapting those identified by (Aikens et al. 2022): (1) What information might flight calls encode, and what can listening individuals learn from these signals? (2) Do individuals respond differently to flight calls over their lifetimes, as the balance shifts between individual experience and social information? (3) How do different migration characteristics, such as distance or complexity, affect how migrants use vocal information? (4) Do species that vocalize during migration show different patterns of migratory evolution? (5) How can vocalization data directly inform conservation and management, for example to lessen the risk of fatal building collisions (Winger et al. 2019)?

Acoustics as a movement ecology tool

Bioacoustics is increasingly important for studying movement ecology. A study of this scope was made possible only through recent advances in machine learning that automate an otherwise laborious detection and identification process. Further work with acoustics promises to reveal more about associations among and within species, as well as the decision-making and conservation status of migratory birds. Our inferences drawn from acoustic monitoring will likely be influenced by factors that impact the vocalization rate of species and individuals, such as environmental conditions, social context, and individual traits (Farnsworth 2005). Currently, it is not possible to distinguish individuals by call with a standard recording setup, which prevented us from investigating associations among conspecifics. However, recent evidence indicates that flight calls may encode individual identity information in at least some species (Ress et al. 2024; Griffiths et al. 2016), which suggests that this may be possible as acoustic analysis methods improve. Distinguishing individuals is currently only possible using microphone arrays that allow calling birds' locations to be triangulated, but this technique requires significant logistical challenges to implement at scale (Gayk and Mennill 2020). Given

our results, we would hypothesize that intraspecific social associations also occur during nocturnal migratory flights (Gayk and Mennill 2023). Finally, it is important to recognize that not all migratory species vocalize during migration (W. R. Evans and O'Brien 2002; Farnsworth 2005). Future work that integrates acoustic data with thermal imagery or small-scale radar data could provide a more holistic understanding of in-flight behavior during nocturnal migration.

Implications

The vocalizations given by birds during migratory flights provide a valuable resource for monitoring the movements and populations of migratory birds, studying their ecologies (Farnsworth 2005), and even understanding anthropogenic hazards like light pollution (Winger et al. 2019). Here, we demonstrate that flight calls provide a window onto a hidden network of interspecific associations. This study highlights the need for further investigation into the social context of animal migration. Recent work supports the important role of transient interspecific relationships during stopover (DeSimone et al. 2024), and we propose that social relationships are also important during migratory flights. Given substantial declines in migratory bird populations (Rosenberg et al. 2019), it is likely that social associations during migration are diminishing, with unknown consequences. Any such density-dependent effects may be complex; a lack of social information might, for example, impede navigational decision-making, impact the duration and energy expenditure of migration, and increase mortality risk (Cohen and Satterfield 2020; Aikens et al. 2022). An understanding of these dynamics is essential to assessing and mitigating negative impacts on populations.

Resource availability

Lead contact

Requests for further information and resources should be directed to and will be fulfilled by the lead contact, Benjamin Van Doren (vandoren@illinois.edu).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- Data have been deposited at Mendeley Data and are publicly available at DOI 10.17632/dxx5khdzjs.1 as of the date of publication.
- All original code has been deposited at Mendeley Data and is publicly available at DOI 10.17632/dxx5khdzjs.1 as of the date of publication.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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Author contributions

BMVD conceived of the study, collected data, performed analysis, and wrote the initial manuscript draft. JAF, FH, and JD shaped the data analysis. All authors contributed to the final written draft of the manuscript.

Declaration of interests

The authors declare no competing interests.

Supplemental information

Document S1. Tables S1-S4 and Figures S1-S4.

Figures

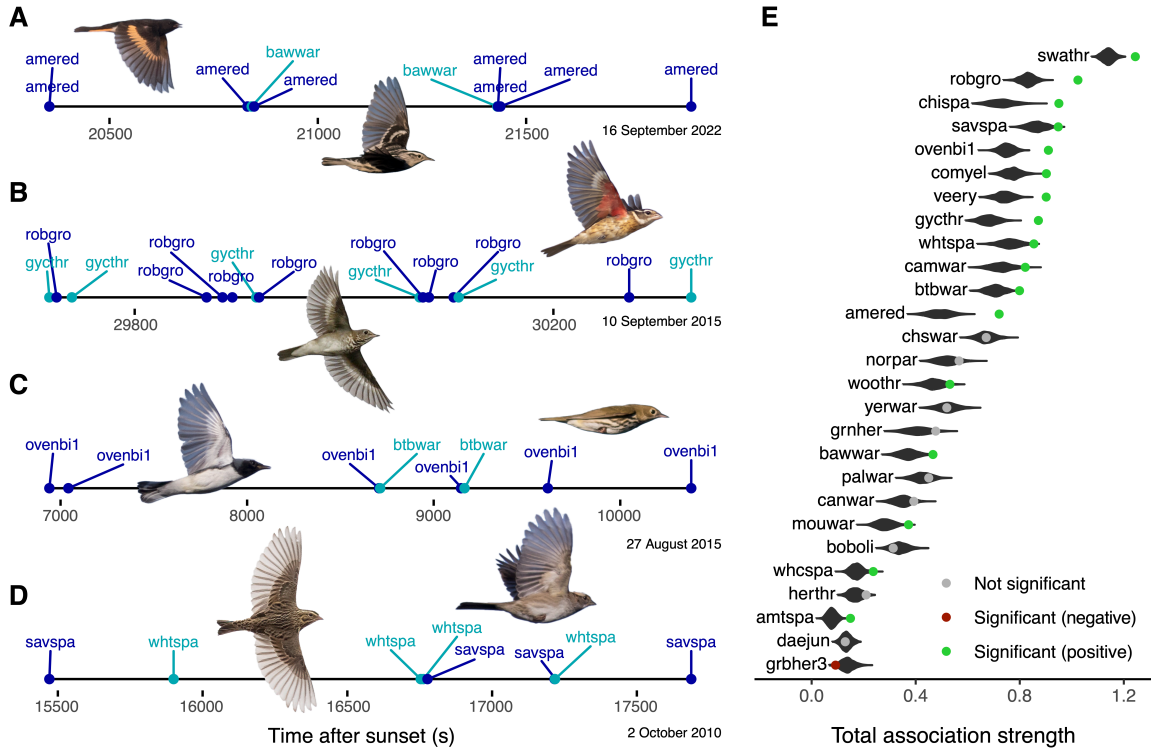


Figure 1: **Example time series of detections during nocturnal migration and tests of social association strength.** Photos from the Macaulay Library at the Cornell Lab of Ornithology. (a) American Redstart (amered; ML112702211) and Black-and white Warbler (bawwar; ML68301071). (b) Gray-cheeked Thrush (gycthr; ML263074761) and Rose-breasted Grosbeak (robgro; ML356438721). (c) Black-throated Blue Warbler (btbwar; ML260441631) and Ovenbird (ovenbi1; ML68440361). (d) Savannah Sparrow (savspa; ML500164831) and White-throated Sparrow (whtsa; ML194675571). (e) Tests of total association strength (weighted degree centrality) for social association network (30-s windows). Violin plots show null distribution and points show observed node strength; statistical significance indicates that a species has a stronger or weaker observed node strength than expected under the null hypothesis. Green points indicate that the observed node strength is significantly greater than expected and that heterospecific attraction can be inferred.

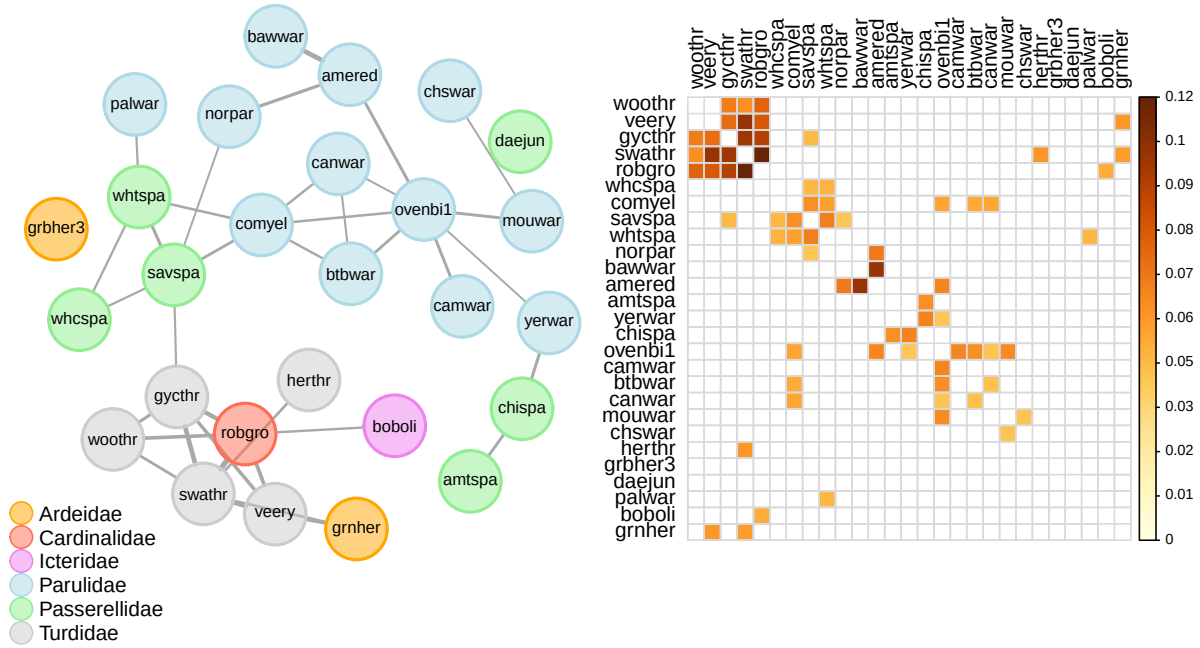


Figure 2: **Significant social associations.** Network diagram and heatmap show only statistically significant edges (30-s time windows). Heatmap values show association strength for each species pair. See Figure S2 and Table S1.

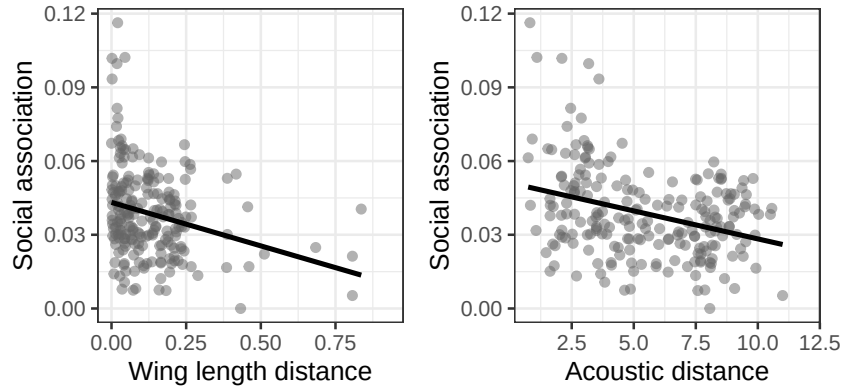


Figure 3: **Scatterplots of statistically significant pairwise species relationships.** Each point represents a species pair. Y-axis represents association strength for each species pair, and x-axes show pairwise phenotype distances. Best linear fit drawn to aid interpretation; refer to matrix correlations for coefficient estimates and statistical significance. Plots shown from data generated with 30-s time windows. See Figure S3 and Table S2.

Tables

Table 1: **Nonparametric matrix correlations for the response variable of social association, based on 30-s time windows.** Each row corresponds to a single-predictor model. See Table S2.

Predictor	Correlation	P-value	No. Taxa
Stopover affiliation index	0.067	0.308	22
Phylogenetic similarity	0.090	0.899	27
Migration overlap	0.187	0.251	27
Non-breeding range overlap	0.131	0.251	27
Wing length distance	-0.252	0.000	27
Migration habitat similarity	0.081	0.162	27
Acoustic distance	-0.266	0.000	27

STAR Methods

Method details

Acoustic data collection

We collected acoustic recordings of autumn nocturnal bird migration (1 August to 7 December) from 26 sites in eastern North America, encompassing 18308.08 hours of monitoring across 379 nights, with an average of 57.3 ± 34.6 SD nights of monitoring per recording station. The recording data come from three monitoring efforts: (Dataset 1) multi-station monitoring in central New York State during fall 2015 (BirdVox-Full-Season (Farnsworth et al. 2022; Van Doren et al. 2023; Cramer et al. 2020)); (Dataset 2) multi-station monitoring in southern New York State during fall 2010-2011 (Van Doren et al. 2015); and (Dataset 3) a 2000-km recording transect across the Appalachian mountain region in eastern North America during fall 2022. Although the hardware differed by monitoring effort, all units were designed and deployed specifically to record migrating birds’ nocturnal flight calls. The likely maximum sampling range of the sensors was 300-600 m above ground level, depending on species call characteristics and ambient conditions (Stepanian et al. 2016; W. R. Evans and Rosenberg 2000; R. Evans and Mellinger 1999).

Acoustic data processing

To extract nocturnal flight calls from audio data, we used Nighthawk, a machine learning tool designed for detecting and classifying nocturnal flight calls (Van Doren et al. 2024). The

Nighthawk core model (Van Doren et al. 2024) is freely available (Van Doren, Mills, and Stafford 2023), and it has been validated on diverse test data, including on the BirdVox-Full-Season dataset (Dataset 1, above) (Van Doren, Mills, and Stafford 2023). Performance on target datasets can be improved by conducting additional model training with the dataset in question, termed “fine-tuning” (Van Doren et al. 2024). We therefore fine-tuned models on Datasets 2 and 3 to maximize model accuracy on those datasets. We manually screened a representative sample of audio data for flight calls and using this dataset to fine-tune Nighthawk (Van Doren et al. 2024). For Dataset 2, we used existing annotations (Van Doren et al. 2015). For Dataset 3, which had not been previously analyzed, we randomly sampled 310 segments of audio each 10 minutes in duration (total 51.7 h; 0.8% of Dataset 3) and screened these for nocturnal flight calls. We then set aside one half of screened data for model fine-tuning and the other half for model validation (Van Doren et al. 2024). That study evaluated multiple fine-tuning approaches; we used the custom batch construction strategy described in that paper since it requires only one epoch of additional training while producing a model that performs very well on target data and original test data.

After fine-tuning, we ran Nighthawk on all data using the freely available Python utility (Van Doren, Mills, and Stafford 2023). For Dataset 1, we used the core model provided in the public package. For Datasets 2 and 3, we substituted in the corresponding fine-tuned model. We used the following important parameters when running the model:

- **–no-calibration:** do not apply default calibration parameters.
- **–threshold 50:** export all detections with a probability score of 0.50 or greater.
- **–ap-mask 0:** do not filter out taxa based on performance on the core Nighthawk test dataset.
- **–tax-output:** export outputs for each taxonomic level independently.

We performed data processing on Amazon Web Services to parallelize inference across thousands of CPU cores. We mapped each detection to time relative to nautical twilight, when the center of the sun is 12 degrees or more below the horizon. We retained detections occurring during the nocturnal period after nautical dusk or before nautical dawn, when any detected flight calls can confidently be attributed to individuals in active nocturnal migratory flight.

Nighthawk returns classifications at multiple taxonomic levels, including order, family, and species. Because we were focused on testing relationships among well-represented species, we only included detections at the species level for species with >250 detections across the dataset. Although our focus is on Passeriformes, we also included two nocturnal migrant species in the order Pelecaniformes to examine the potential for associations between passerines and other orders.

Because our analysis relies on high quality detection data, we used a multi-step review process to ensure detection accuracy. First, we randomly sampled up to 200 detections per species per dataset and manually screened these detections for accuracy. We used the results to set confidence thresholds for each species in each dataset to target a precision of approximately 0.95 on all classes. After retaining detections with confidence scores above the corresponding

thresholds, authors BMVD and AF manually reviewed all acoustic detections from the subset of 30-s time windows that included multiple taxa. In other words, we manually reviewed all data that contributed to any potential associations among species pairs. In total, we reviewed 64909 detections. We conservatively removed all detections with any ambiguity in species identity, primarily recordings with a low signal-to-noise ratio. In total, we removed 6538 detections (10.1% of those reviewed). After all filtering steps, our acoustic dataset comprised 177962 detections of flight calls from 27 species (Figure 1).

Network generation: Fine co-occurrence networks

We used acoustic detections to construct networks of observed species co-occurrence in the acoustic temporal data stream. In these networks, stronger connections among species indicate that those species were recorded together more frequently during migratory flights. To construct networks, we split our acoustic data streams into independent, nonoverlapping 30-s windows and grouped species detected in the same 30-s window into “events.” Although this choice was somewhat arbitrary, an interval of 30 s corresponds to a maximum linear distance of 450 m, assuming a bird’s groundspeed of 15 ms^{-1} ; we reasoned that migrating birds recorded in the same 30-s window would likely be close enough to hear one another and potentially exchange information. We quantified network connections from these 30-s events using the default Simple Ratio Index formula implemented in the `get_network` function in the R package `asnipe` (Farine 2013). For a given pair of species, the Simple Ratio Index is calculated by dividing the number of events (i.e. 30-s windows) in which both species occur by the number of events in which either one or both species occur. We also constructed networks using 15-s and 60-s windows to assess whether the results were sensitive to the choice of window length. Networks generated from different window sizes were very tightly correlated using Mantel correlations (30-s vs. 60-s: $r = 0.99$; 30-s vs. 15-s: $r = 0.99$). We refer to these networks as **fine co-occurrence networks** because they capture the degree to which vocalizations of each species pair occur close together in our data stream.

Network generation: Coarse co-occurrence network

The network connection strength among species in fine co-occurrence networks is partly a function of species’ similarity in migration timing, geographic distributions, and other factors unrelated to social associations. We accounted for this by constructing an acoustic network as described above, but with events defined using longer 15-*minute* time windows. Rather than considering fine-scale social associations, this **coarse co-occurrence network** captures broader species co-occurrence in the dataset driven by shared seasonal timing, geography, and consistent behavioral patterns over the nocturnal period.

Network generation: Social association networks

Because connections among species in fine co-occurrence networks may arise from factors that are unrelated to species' propensity to actively associate, we used the coarse co-occurrence network to control for these factors. The goal was to generate networks that explicitly captured the degree to which species' vocalizations occurred synchronously, independent of shared timing, geography, or other non-social factors. We calculated **social association networks** as follows: for each species pair, we subset the data to only the 15-minute time periods in which both species were detected. Then, we calculated the Simple Ratio Index on this subset using 30-s windows as described above. To ensure that our measures were reliable, we did not calculate social association for species pairs for which there were less than 100 15-minute windows in which the two species occurred (i.e. <100 association opportunities). After performing these calculations for all pairs of species, the resulting social association network captured the degree to which vocalizations of each species pair occur close together, *conditioned* on the time periods during which both species are detected. Because this metric is conditioned on species co-occurrence, these networks do not depend on seasonal migration timing or nocturnal vocalization patterns; they only quantify the degree of acoustic synchronicity among species pairs independent of broader temporal or geographic patterns. As above, we also generated social association networks for 15-s and 60-s window durations to assess whether our results were sensitive to the choice of window length. Networks generated from different window sizes were very tightly correlated using Mantel correlations (30-s vs. 60-s: $r = 0.93$; 30-s vs. 15-s: $r = 0.94$).

Generating network covariates

To test hypotheses about the drivers of species associations during migration, we generated seven covariates that summarize the similarity of each species pair in phylogeny, spatiotemporal distribution, habitat preferences, social relationships during stopover, morphology, and vocalization structure.

Phylogenetic relationships

We obtained a phylogenetic tree of the species included in our study using the R package `clootl` (McTavish et al. 2024). We used the `extractTree` function in that package to output a tree and used the `cophenetic.phylo` function in the R package `ape` (Paradis and Schliep 2019) to convert the tree topology to pairwise phylogenetic distances for all species pairs. We used the inverse of these distance values as measures of phylogenetic similarity.

Species geographic ranges

For each species pair, we calculated pairwise range overlap scores for their non-breeding ranges. We used species ranges modeled by eBird Status & Trends (Fink et al. 2020) to calculate

pairwise overlap for each species pair. We used the eBird Status & Trends models accessible in the R package `ebirdst` (v. 2.2021.3) (Strimas-Mackey et al. 2022). We downloaded Status & Trends data for each species and used the `load_ranges` function to extract the modeled ranges. We then calculated the range overlap for each species pair by dividing the area of the intersection of the two ranges by the area of the union of the two ranges.

Migration overlap

We estimated the overall migration similarity for each species pair using a spatiotemporal measure of overlap in the species' geographic distribution during migration season. We extracted weekly 27x27 km relative abundance rasters for each species using the `ebirdst` package and subset these to the migration period for that species as defined by eBird in the package. For each species pair, we found the total number of cells where modeled relative abundance was greater than zero for both species, and we divided this by the total number of raster cells where relative abundance was greater than zero for either species. This resulted in a proportion of overlapping cells for each week. Finally, we took the mean weekly overlap proportion across all weeks of the migration periods. This resulted in a single proportion value for each species pair that captured the spatiotemporal overlap in their geographic distributions during the migration period.

Stopover habitat

To calculate the degree of similarity in the habitat preferences of each species pair during the migration season, we extracted weekly habitat associations from eBird Status & Trends data (Fink et al. 2020). We filtered habitat association data to the migration period for each species using the migration period dates provided in the `ebirdst` package. Using all available habitat association characters, we used the `dist` function in R to calculate a pairwise distance matrix that captured the overall pairwise similarity in habitat associations for all species pairs.

Social affiliations during stopover

To assess migratory species' social networks at stopover sites, we used over half a million records of banded migratory birds collected during spring and fall migration seasons by Braddock Bay Bird Observatory (43.324, -77.717), Long Point Bird Observatory's banding stations at Old Cut (42.584, -80.398) and Breakwater (42.561, -80.284), Powdermill Avian Research Center (40.164, -79.268), and Michigan State Bird Observatory's Burke Lake banding station (42.812, -84.383). More details about these datasets are reported in (DeSimone et al. 2024).

Following that study (DeSimone et al. 2024), we calculated species associations from the banding data using the Simple Ratio Index. Next, we calculated generalized affiliation indices by regressing the species associations against measures of temporal overlap, spatial overlap, and relative abundance. The standardized residuals of the regression are the generalized affiliation indices for each species pair. The affiliation indices quantify the degree to which

two species associate after accounting for structural features of the data, including temporal overlap, spatial overlap, and relative abundance. We calculated fall affiliation indices separately for each site and averaged affiliation values across sites. We included only species with >100 fall captures.

Wing-length measurements

Because body morphology impacts flight behavior and could contribute to in-flight dynamics, we extracted wing-length measurements from the AVONET dataset (Tobias et al. 2022) for each species. Wing length is associated with flight speed and flight style and may influence species' in-flight associations. For each species pair, we calculated the Euclidean distance between the base-10 logarithms of their wing lengths as a measure of the difference in wing length (hereafter “wing length distance”).

Acoustic distance

It is possible that bird species with more acoustically similar flight calls may be more likely to interact during migration (Gayk and Mennill 2023). To evaluate this hypothesis, we calculated the acoustic distance of the vocalizations given by species in our dataset. We randomly sampled 200 vocalizations for each species from the expert-verified set of recordings used in Van Doren et al. (2024) and selected recordings with sufficiently clean spectrograms for further analysis. We retained a mean of 61.1 ± 16.8 SD (range 13–89) vocalizations per species. We used Raven Pro 1.6 (K. Lisa Yang Center for Conservation Bioacoustics 2024) to manually draw bounding boxes around each call and used the `spectro_analysis` function in the R package `warbleR` (Araya-Salas and Smith-Vidaurre 2017) to extract a series of 26 spectrographic measurements. See `warbleR` documentation for a description of measurements. We summarized these measurements using a Principal Component Analysis (PCA function in R package `FactoMineR` (Lê, Josse, and Husson 2008)) and extracted the first 5 components, comprising 89.7% of total variance. We used the centroids of each species in multidimensional PCA space to generate a distance matrix (`dist` function in the base R package `stats` (R Core Team 2024)) that describes acoustic distance among species. Smaller values indicate more similar vocalizations.

Quantification and statistical analysis

Generating null network distributions with permutations

To test the statistical significance of network parameters, including the strength of species connections in co-occurrence and social association networks, we generated null distributions of network parameters using custom permutations of the original data stream. The permutation procedure was as follows: first, we divided acoustic detection data into 15-minute periods for each site and date. Then, *for each species* in each 15-minute period, we shifted the timing of

all detections by a random time interval between 0–15 minutes. Each species present in the 15-minute period was shifted by a different random interval, and all calls of that species in that period were shifted by the same amount. If the procedure shifted any detections further than the bounds of the 15-minute period, those detections were “wrapped around” to the beginning of the period. In this way, each permuted time period maintained the same quantity and the same temporal structure of vocalizations of each species as the original dataset. This procedure randomly changed the degree to which different species’ vocalizations occurred relative to other species, allowing us to test a null hypothesis of no association among species in vocalization patterns. After applying this permutation procedure independently to every 15-minute period in the dataset, we calculated co-occurrence and social association networks from the permuted data using the procedures described above. We repeated this procedure 1000 times, yielding 1000 null networks for 15-, 30-, and 60-s window sizes.

Testing for network randomness

We first evaluated whether networks of co-occurrence and social association differed significantly from random. We calculated the network coefficient of variation by taking the standard deviation of the adjacency matrix and dividing it by the mean of the adjacency matrix. We performed this calculation for observed networks and for all permuted networks. If the observed coefficient of variation was greater than the 0.95 quantile of the corresponding null (permutation) distribution, we considered the network non-random at the $P < 0.05$ level. Networks contained $n = 27$ species.

Testing for social associations among species

We evaluated the statistical significance of each species’ ($n = 27$) connections with other species in networks using null distributions derived from the permuted networks. For each species, we quantified its overall tendency to occur with other species during migratory flights by summing the strength of all network connections between the focal species and other species, also known as the weighted degree centrality. Larger degree values indicate that a species shows stronger and/or more numerous connections to other species in the network. We compared total association strength values calculated from observed co-occurrence and social association networks to those calculated from the corresponding permuted networks. We considered a species to show statistically significant associations with other species if the observed total association strength for that species was greater than the 0.95 quantile of the corresponding null distribution derived from the permuted networks.

We assessed statistical significance for every species pair in co-occurrence and social association networks using the same procedure: we compared the connection strength for a given species pair with the null distribution of values derived from the corresponding null networks. We again assessed significance by comparing observed values to the corresponding null distribution.

We corrected p-values for multiple testing using a false discovery rate correction with a false discovery rate of 0.05.

Explaining migrant associations

Finally, we tested whether in-flight associations among species could be explained by phylogeny, spatiotemporal distribution, habitat preferences, social relationships during stopover, morphology, or vocalization structure. We constructed single-predictor statistical models in which the response variable was social association. As described above, social association does not depend on seasonal migration timing or nocturnal vocalization patterns; it quantifies the degree of social association among species pairs independent of broader temporal or geographic patterns.

We evaluated statistical significance using a modification of the Mantel test procedure (`mantel` function in R package `vegan` (Oksanen et al. 2024)): for each predictor, we calculated the Mantel matrix correlation between that predictor and the social association matrix; then, we compared this observed statistic to the null distribution of test statistics obtained from our custom set of permuted social association networks. For each test, the p-value was the proportion of permuted networks that achieved a Mantel correlation equal to or more extreme than the observed statistic. To eliminate any bias from skewed data distributions, where outliers could exert a strong influence on the correlation value, we calculated Mantel statistics using the nonparametric Spearman rank correlation. For comparison, we also obtained results using the standard Pearson correlation. For the single-predictor case, the Pearson-based tests of statistical significance were equivalent to those obtained using Multiple Regression Quadratic Assignment Procedure (MRQAP) to regress predictor matrices on the response matrix, as recommended for networks (Farine 2017), using the `mrqap.custom.null` function in the R package `asnipe` (Farine 2013). The number of taxa in each model ranged between 22–27 and is reported for each model in Table 1.

We elected to use a series of single-predictor models instead of multiple matrix regression for the following reasons: first, we did not have stopover affiliation data for all species, and this imbalance would have required removing those species from a multiple regression model and/or running multiple sets of models; second, we wanted to avoid collinearity among predictor variables from biasing coefficient estimates; third, we found matrix correlation statistics, which vary between -1 to 1, to be more easily interpretable than multiple regression coefficients, which are unbounded; and fourth, this allowed us to test our hypotheses using more robust nonparametric rank correlations.

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